TELCO CUSTOMER CHURN PREDICTION

Machine Learning Classification Project

Dataset	Telco Customer Churn (WA_Fn-UseC_-Telco-Customer-Churn.csv)
Task	Binary classification: predict whether a customer churns
Models	Logistic Regression, Decision Tree, SVM, K-Nearest Neighbors, Soft Voting Classifier
Highest reported accuracy	**82.04% — Soft Voting Classifier**

This report presents the project workflow, model evaluation, supplied visualizations, interpretation of findings, limitations, and recommendations.

1. Executive Summary

This project analyses customer churn using the Telco Customer Churn dataset and compares five classification models. The supplied script reports 7,043 records and 21 original columns. After preprocessing and an 80/20 train–test split, the models are assessed using accuracy, precision, recall, F1-score, cross-validation, and precision–recall curves.

The Soft Voting Classifier has the highest recorded test accuracy at 82.04%. Logistic Regression is close at 81.83% and has the highest reported recall (59.25%) and F1-score (63.32%) among the five models. The Decision Tree is less effective at identifying the positive class, with recall of 32.44%. Therefore, accuracy should not be the only measure used to assess a churn model.

2. Problem Statement and Objectives

Problem: Customer churn can reduce recurring revenue and increase customer-acquisition costs. The task is to classify customers as churned or not churned using available customer and service attributes.

Objectives: Prepare the data, train and compare classification models, interpret evaluation charts, and recommend a model based on the balance between overall accuracy and the ability to identify churners.

3. Dataset and Preprocessing

The source script loads *WA_Fn-UseC_-Telco-Customer-Churn.csv* and reports 7,043 rows × 21 columns. It checks missing values and duplicates, converts *TotalCharges* to numeric, and fills unparseable values with zero. Descriptive statistics are calculated for SeniorCitizen, tenure, MonthlyCharges, and TotalCharges.

Categorical variables are one-hot encoded; Contract is ordinal-encoded in the order Month-to-month, One year, Two year; and Churn is label-encoded. The customerID column is dropped. Data is split into 80% training and 20% testing with random_state=42, and features are standardized using a scaler fitted on the training set.

Methodology note: Filling unparseable charges with zero may influence results. A more robust approach is to handle missing values explicitly and fit preprocessing steps inside a scikit-learn Pipeline during cross-validation.

4. Model Comparison and Evaluation

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	81.83%	68.00%	59.25%	63.32%
Decision Tree	77.57%	64.02%	32.44%	43.06%
SVM	81.76%	68.01%	58.71%	63.02%
K-Nearest Neighbors	79.91%	63.64%	56.30%	59.74%
Soft Voting Classifier	82.04%	70.13%	56.03%	62.30%

Metric guide. Accuracy is the fraction of all test examples classified correctly. Precision measures how many predicted churners were actual churners. Recall measures how many actual churners were detected. F1-score balances precision and recall. For churn management, false negatives matter because customers at risk may not be identified.

5. Data Story and Key Findings

1 — Ensemble accuracy is highest. Soft Voting reaches 82.04%, about 0.21 percentage points above Logistic Regression. This is a modest improvement rather than a dramatic performance gap.

2 — Logistic Regression has the strongest balance in the reported metrics. It records 59.25% recall and 63.32% F1-score. The ensemble has higher precision (70.13%), but lower recall (56.03%) and F1-score (62.30%).

3 — Decision Tree misses more churners. Its recall of 32.44% is substantially below the other models. Its 77.57% accuracy should not hide this weakness when identifying churners is the priority.

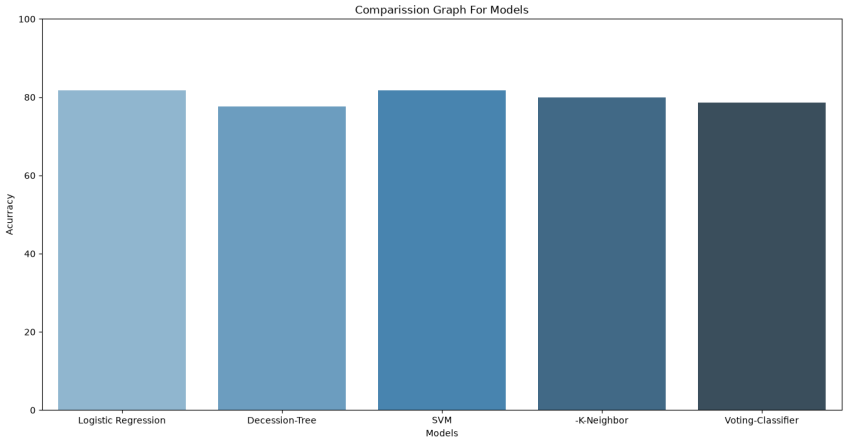
4 — Tenure relates strongly to total charges. The correlation heatmap reports a correlation of 0.83 between tenure and TotalCharges, and 0.65 between MonthlyCharges and TotalCharges. Correlation does not establish cause.

5 — Precision–recall curves add context. The supplied chart legends show average precision (AP) around 0.68 for Logistic Regression, 0.67 for SVM, 0.60 for KNN, and 0.48 for Decision Tree. AP summarizes performance across thresholds and is not the same as precision or recall at one threshold.

6. Graph Gallery and Interpretation

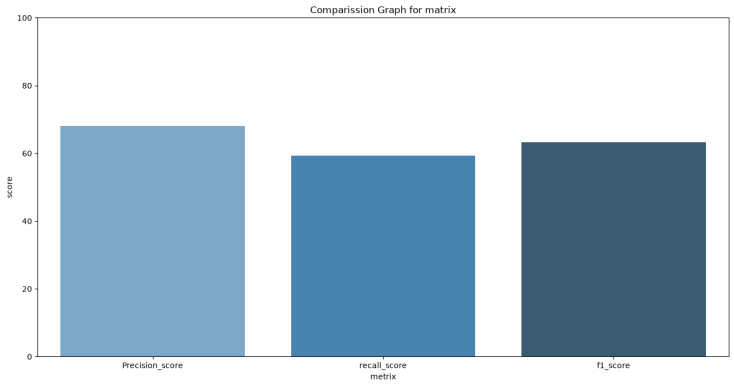
The following figures are the chart images supplied with the project. Values are interpreted from the charts and the model results reported in the supplied script.

Model accuracy comparison



The Soft Voting Classifier has the highest reported test accuracy (82.04%), narrowly ahead of Logistic Regression and SVM.

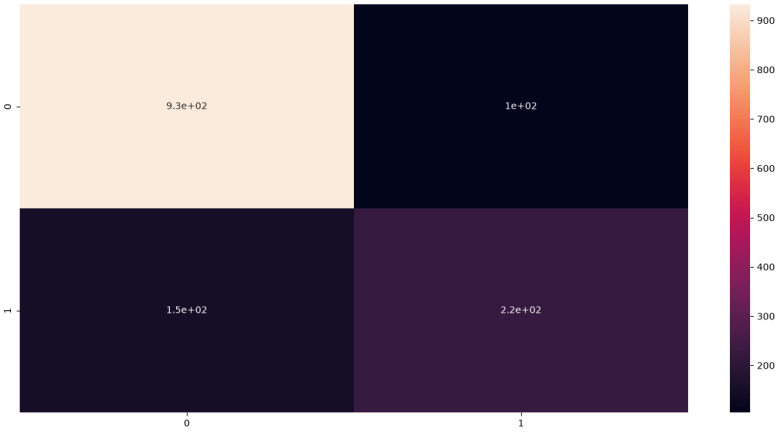
Classification metrics: Logistic Regression



The supplied chart shows precision around 68%, recall around 59%, and F1-score around 63%.

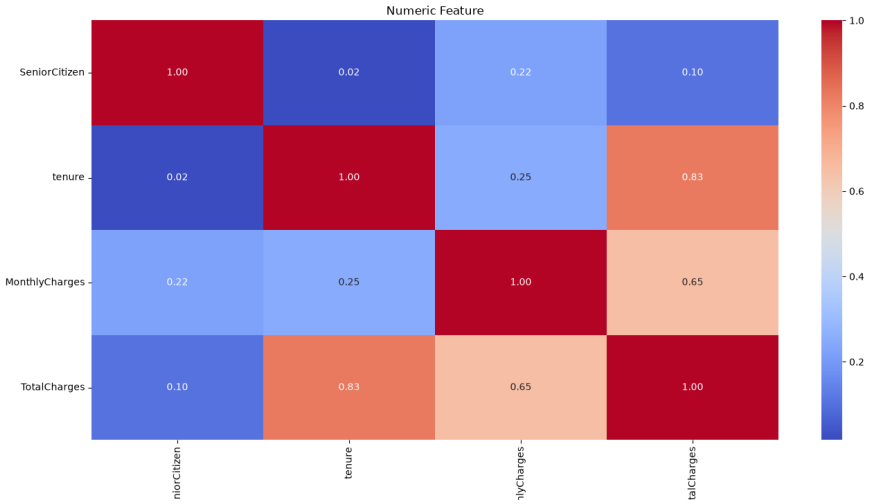
Graph Gallery and Interpretation (continued)

Confusion matrix: Logistic Regression



The displayed matrix is approximately [[930, 100], [150, 220]] for classes 0 and 1. Confirm the class-label order and axis meanings before presenting.

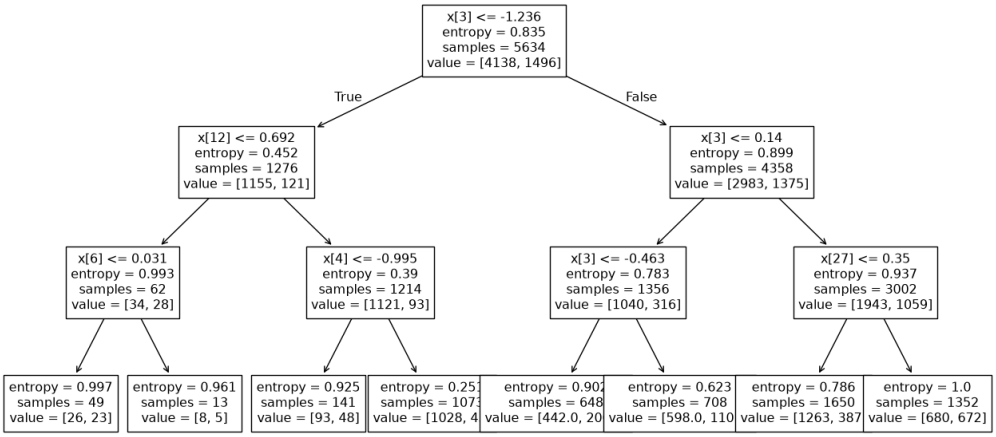
Numeric-feature correlation heatmap



Tenure and TotalCharges show a strong positive correlation (0.83); MonthlyCharges and TotalCharges are also positively correlated (0.65).

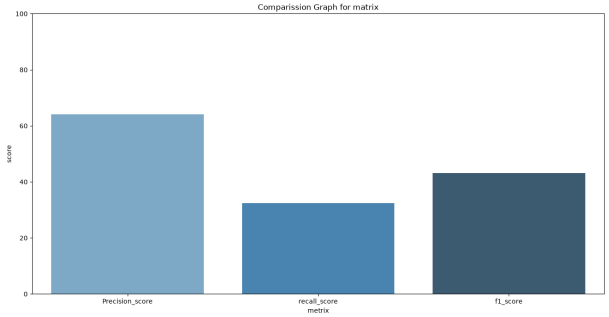
Graph Gallery and Interpretation (continued)

Decision-tree visualization



The tree visualizes decision rules and class counts. Split labels are encoded feature indices rather than human-readable feature names, so they should be mapped to the feature matrix before interpreting individual rules.

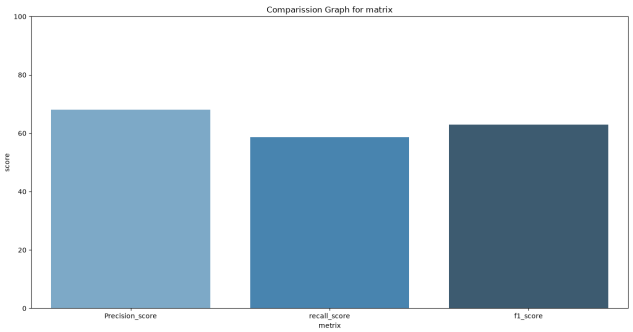
Classification metrics: Decision Tree



The reported precision is about 64.02%, recall about 32.44%, and F1-score about 43.06%.

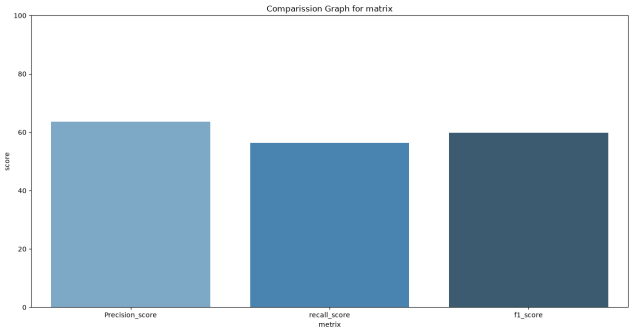
Graph Gallery and Interpretation (continued)

Classification metrics: SVM



The reported precision is about 68.01%, recall about 58.71%, and F1-score about 63.02%.

Classification metrics: K-Nearest Neighbors

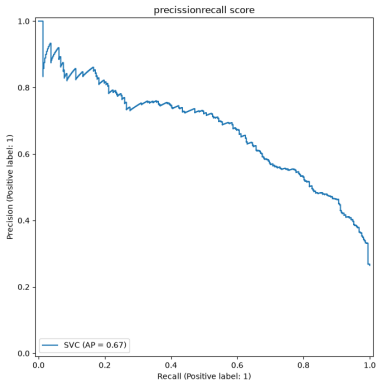


The reported precision is about 63.64%, recall about 56.30%, and F1-score about 59.74%.

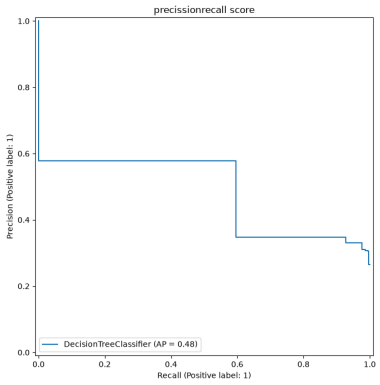
Precision–Recall Curves

Precision–recall curves show the trade-off between precision and recall as the decision threshold changes. They are particularly useful when the positive class is less common or when false negatives and false positives have different costs.

SVM — average precision approximately 0.67

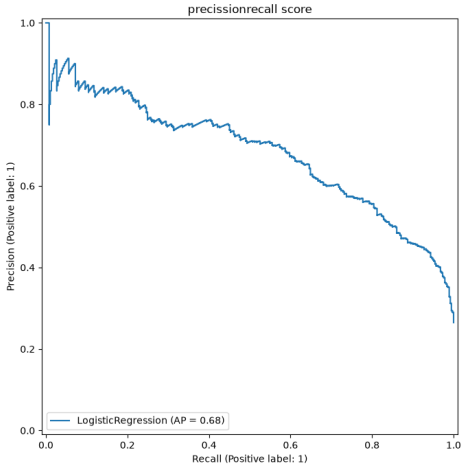


Decision Tree — average precision approximately 0.48

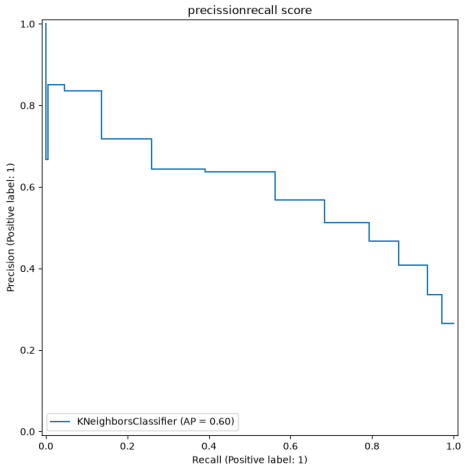


Precision–Recall Curves (continued)

Logistic Regression — average precision approximately 0.68



K-Nearest Neighbors — average precision approximately 0.60



7. Limitations and Recommendations

• The source script uses an absolute Windows file path. Update it before running the project on another computer.

• Put preprocessing and encoding steps into a scikit-learn Pipeline to reduce leakage risk and improve reproducibility.

• Some cross-validation calls use default estimators rather than the final tuned configurations. Use consistent validation for the final model settings.

• Review the final feature matrix and confirm that only intended predictors remain. Inspect X.columns before fitting.

• Use stratified splitting and report class-wise metrics. If churn recall is the priority, explore threshold tuning and class weights using validation data.

• State the positive class and confusion-matrix axis order clearly in the presentation.

8. Conclusion

The supplied experiment suggests that Logistic Regression, SVM, and Soft Voting perform similarly in overall accuracy. Soft Voting ranks first in accuracy, while Logistic Regression has the highest reported recall and F1-score among the evaluated models. The final choice should depend on whether the business objective emphasizes overall correct predictions, reducing false alarms, or detecting as many churners as possible.

9. Reproducibility

Install Python and the dependencies pandas, numpy, matplotlib, seaborn, and scikit-learn. Place the CSV dataset in an accessible location, update the path passed to *pd.read_csv* in *Project 2.py*, and run the script with Python. Some plotting sections in the original source are commented out; uncomment them if you need to regenerate the corresponding charts.

Report scope: This PDF contains the project explanation, results, graphs, and interpretation. The full source-code appendix has been intentionally omitted, as requested.